Four nonanomalous positions: 3-30 Hz

Each position uses its own fitted modal model and input/output delays. Dense analytical evaluation; no old-baseline curve.
The anomalous z = -1 mm record is excluded from this focused comparison.

z = -0.5 mm z = 0 mm z = 0.5 mm z = 1 mm

$u_1 \to y_1$ $u_2 \to y_1$ $u_3 \to y_1$ $u_4 \to y_1$
$u_1 \to y_2$ $u_2 \to y_2$ $u_3 \to y_2$ $u_4 \to y_2$
$u_1 \to y_3$ $u_2 \to y_3$ $u_3 \to y_3$ $u_4 \to y_3$
$u_1 \to y_4$ $u_2 \to y_4$ $u_3 \to y_4$ $u_4 \to y_4$

Magnitude (dB, recorded output / motor-command unit)

Frequency (Hz)